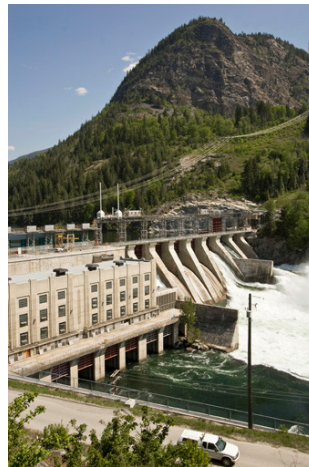


Stevens®

Water Monitoring Systems, Inc.

*Monitoring the Earth's Water
Resources Since 1911*



Improved Water Supply Forecasts Using Soil Moisture Data

4th International Symposium on Soil Water Measurement using Capacitance,
Impedance and Time Domain.

Keith Bellingham P. H.
Stevens Water Monitoring Systems, Inc.

Jolyne Lea P. H.
United States Department of Agriculture
National Resource Conservation Service.



Water Uses in the Western United States

- About 80% of the water consumed is used for irrigation.



Irrigation Districts and Water Distribution



Hydroelectric Power

- Washington 80%
- Oregon 45%
- California 7%
- Idaho 70%
- About 7 to 9 % in the US Nation Wide



Environmental impacts from the consumption of natural waters.

From Irrigation Withdrawals and Return Flows

- Temperature loading
- Salinity Loading
- Flow Conditions
- Total dissolved solids and suspend loads

Dams

Flow Conditions.

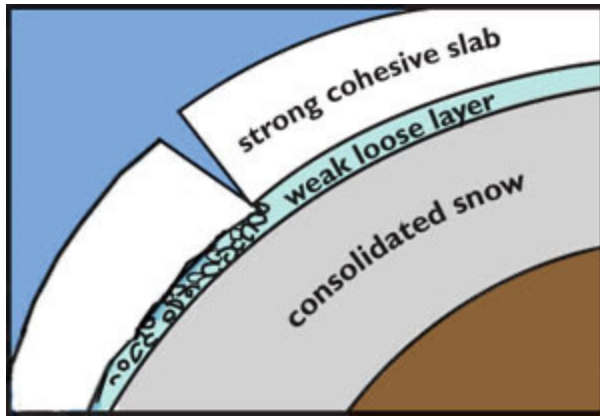


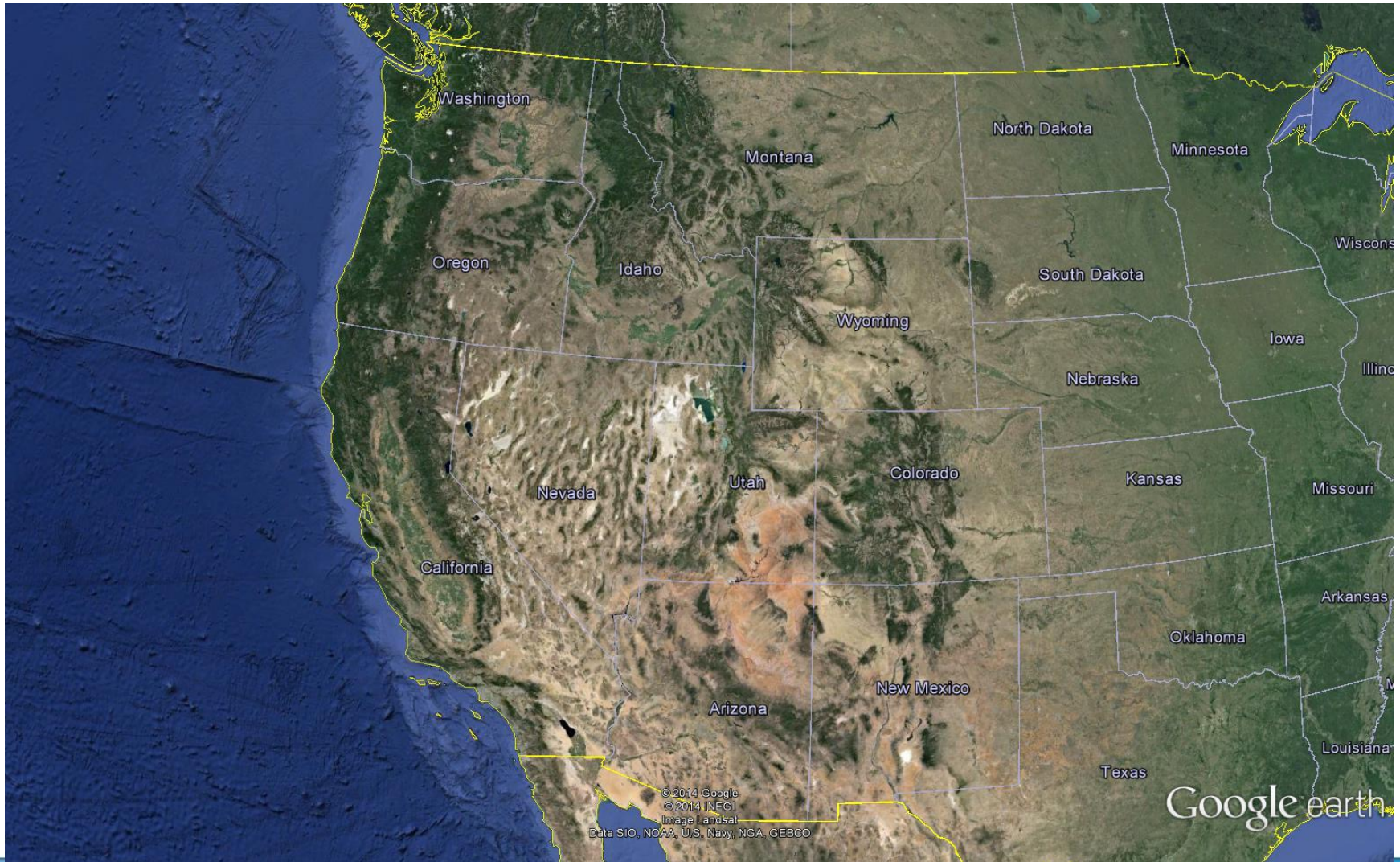
Forecasting the Available of Water

Major Economic Drivers

- Flood Risks
- Energy Trading
- Water for irrigation and food production

Snow Pack





SWE

Snow Water Equivalent

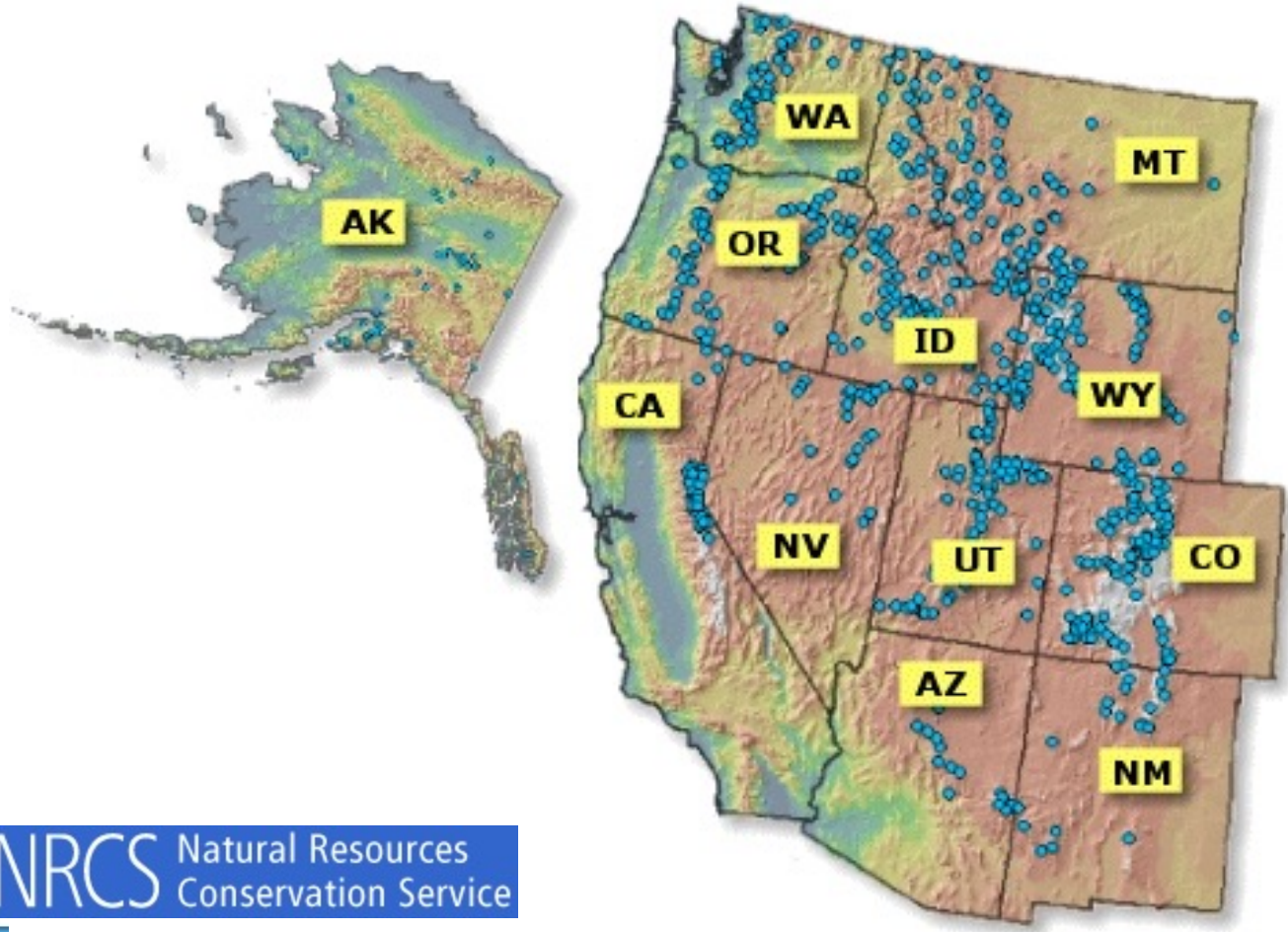


Dr. James Church from the University of Nevada

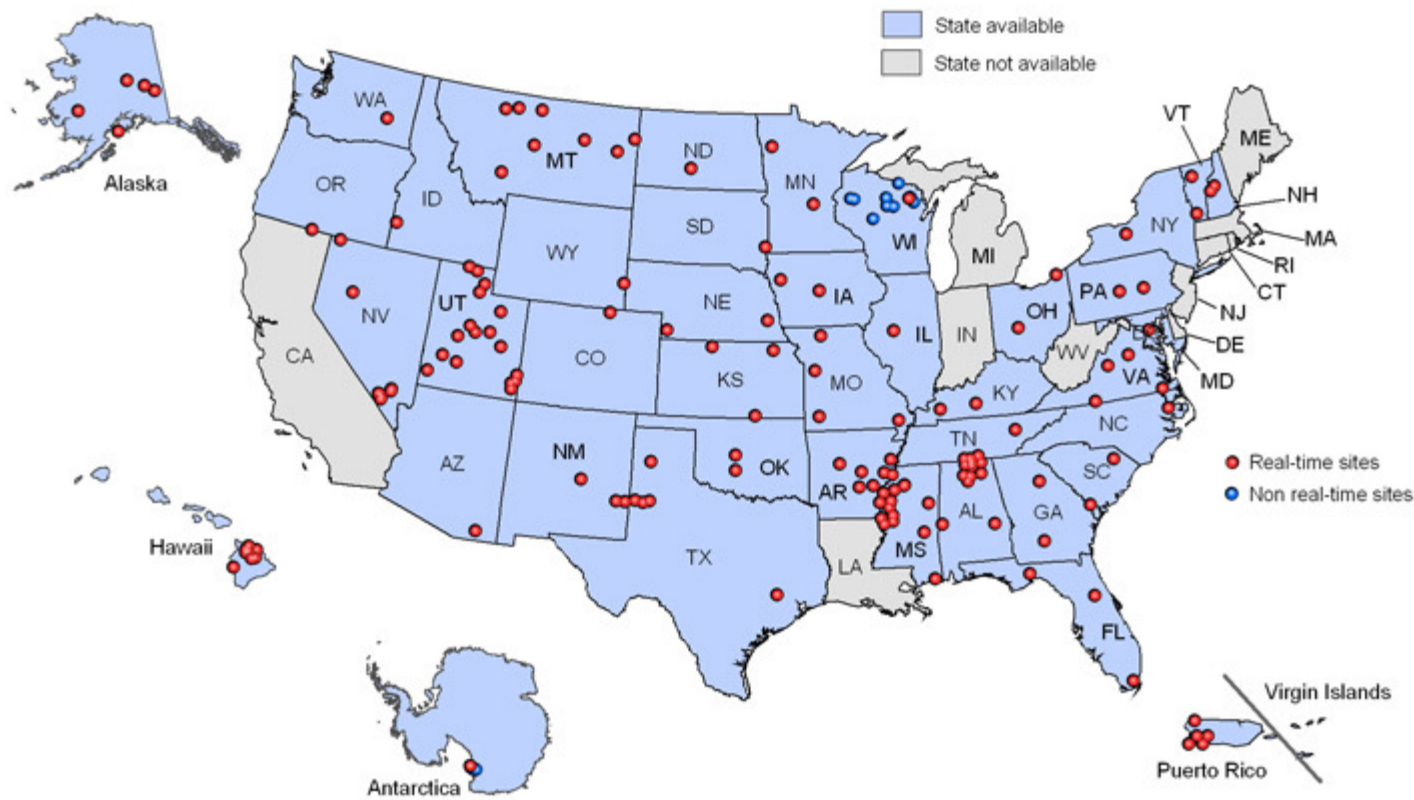
- Dr. James Church in 1906 was the first scientist to study the relationship between stream flow and the snowpack in the western United States.
- He made the first Snow Courses.



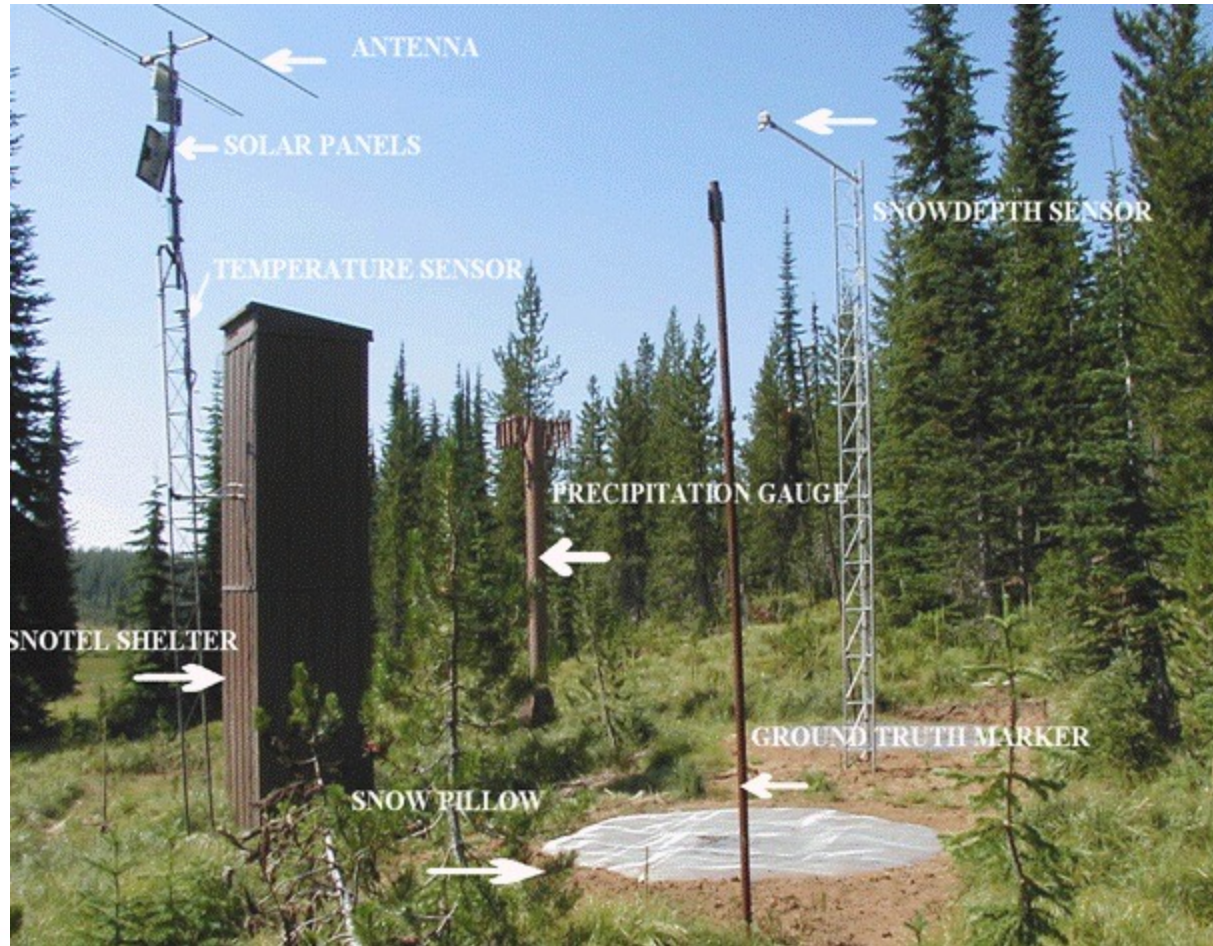
The SNOTEL Network is about 900 sites in 12 states.



USDA SCAN Network



Components of a SNOTEL





The Stevens Hydra Probe used in SNOTEL and SCAN



Forecast Modeling

Statistical Principle Component Regression
(Simplified Equation)

Stream Flow Total May to July ($\times 10^6 \text{ m}^3$) =

$$AX_1 + BX_2 + CX_3 + DX_4 + EX_5 + F$$

Where A, B, C, D, E, and F are coefficients from the eigenvalue of the correlation matrix.

X_1 Antecedent Stream Flow from previous years (10^6 m^3)

X_2 Fall and winter Precipitation (mm)

X_3 Spring precipitation (mm)

X_4 High altitude snow water equivalent (cm)

X_5 Low altitude snow water equivalent (cm)

Deschutes R at Benham Falls nr Bend (2)

14064500: Deschutes R at Benham Falls nr Bend (2)

Group: Middle Columbia Basin
[State Basin Outlook Report](#)

State: Oregon
[USGS Data](#)

Forecast volume in 1000's of acre-feet

	(Drier) -----	Chance of exceeding -----				(Wetter)
Target (71-00 norm)	9 in 10	7 in 10	5 in 10	3 in 10	1 in 10	
MAY-JUL (270)	255	270	280	290	305	
MAY-SEP (445)	425	445	460	475	495	

9 in 10, 255 means that there is a 90% chance that you will have more than 255,000 acre-feet go past the gage. (270) is the 30 year average.

MAY-JUL	94	100	104	107	113
MAY-SEP	96	100	103	107	111

A footnote of "(2)" indicates that the value is natural volume - actual volume may be affected by upstream water management.

Forecast issue date (yyyymmdd): 20120501
Data file creation date : 20120719
Google Earth kml creation date: 20120719

Directions: [To here](#) - [From here](#)

The soil is a significant reservoir for water in a watershed.

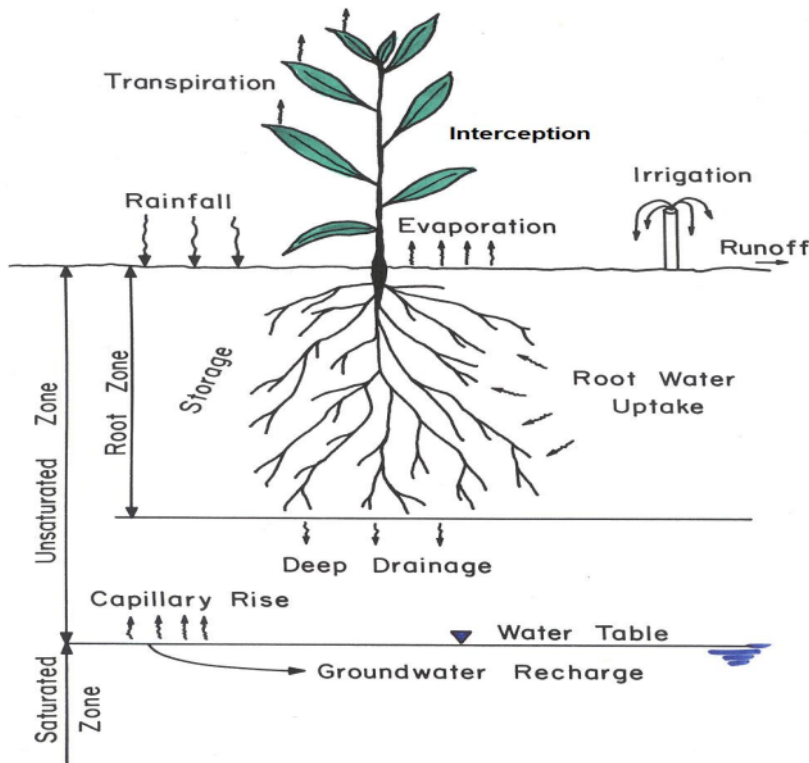


Figure 5.1 Transient water flow processes in soil (Šimůnek and van Genuchten, 2007).

Put it in Prospective

- 270,000 acre feet from to July (332 million cubic meters)
- If a 100 square km watershed has a porosity of 0.5 with one meter to the bedrock, draining from saturation to 25% will yield 24 million cubic meters of waters.



Streamflow forecasts using soil moisture data

Can we use soil moisture data to Improve the Stream Flow Forecast?

Statistical Principle Component Regression

(Simplified Equation)

Stream Flow Total May to July ($\times 10^6 \text{ m}^3$) =

$$AX_1 + BX_2 + CX_3 + DX_4 + EX_5 + FX_6 + H$$

Where A, B, C, D, E, G and H are coefficients from the eigenvalue of the correlation matrix.

X_1 Antecedent Stream Flow from previous years (10^6 m^3)

X_2 Fall and winter Precipitation (mm)

X_3 Spring precipitation (mm)

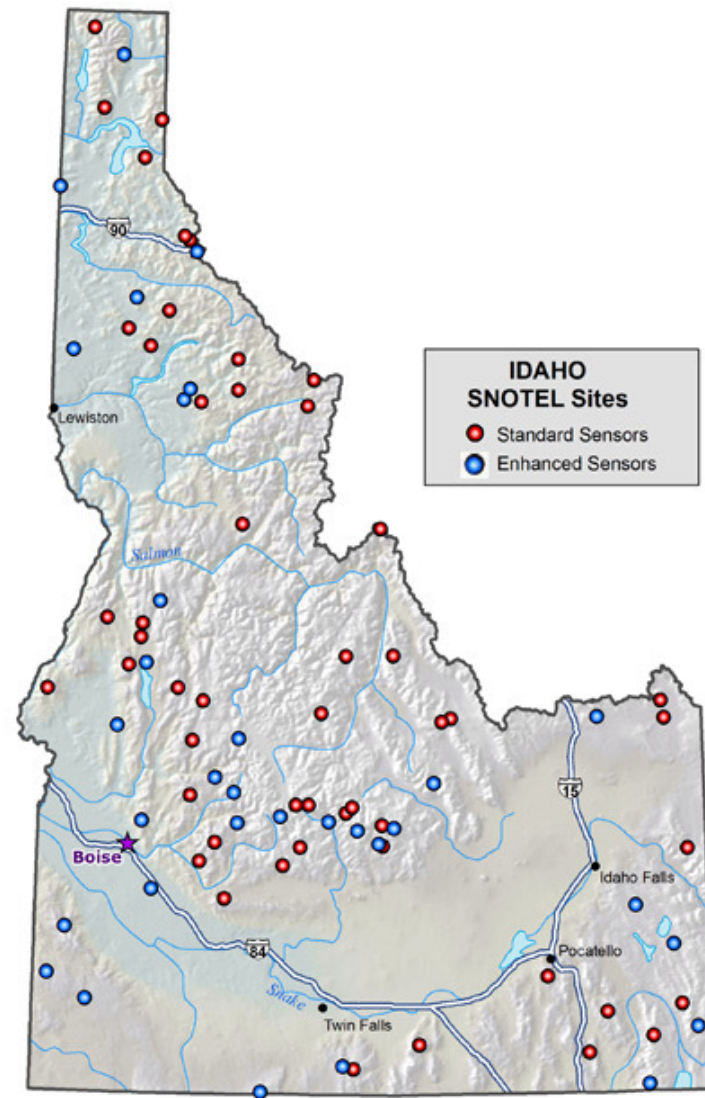
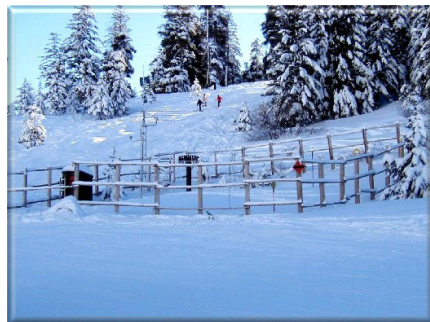
X_4 High altitude snow water equivalent (cm)

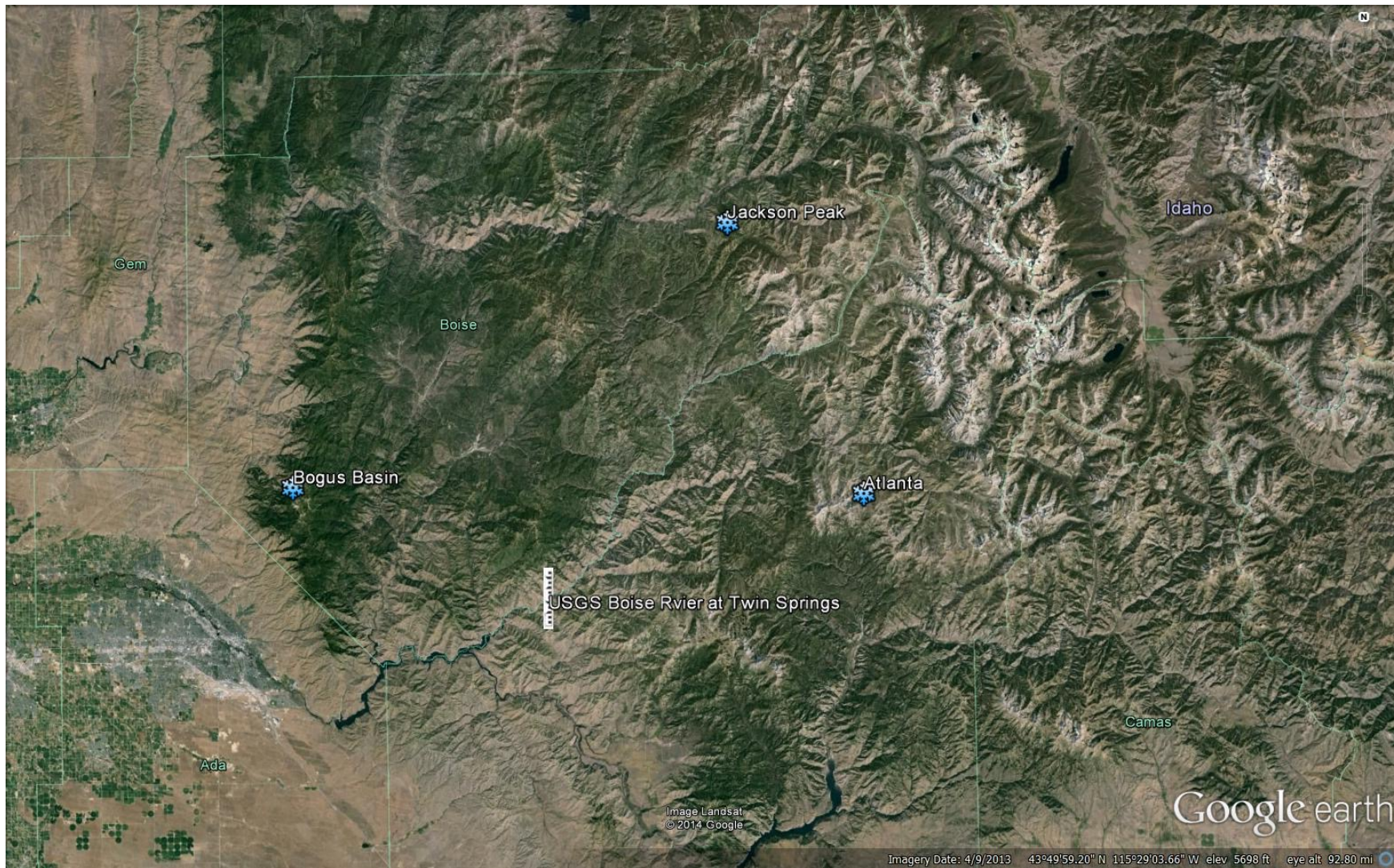
X_5 Low altitude snow water equivalent (cm)

X_6 Soil Moisture Deficit Index From the measure soil moisture data.

- The meteorological data along with the soil moisture data was modeled to predict the seasonal flow at the Boise River. The simulations with the soil moisture data were compared to the historical forecasts to the actual stream flow.

Jackson Peak, Atlanta Summit, and Bogus Basin





SCAN and SNOTEL Use the Stevens Hydra Probe



The Hydra Probe is an Impedance Based Sensor.

Separates the Real from the Imaginary at 50 MHz.
Characterizes the amplitude of the reflected and incident signals ratiometrically to eliminate inter sensor variability.

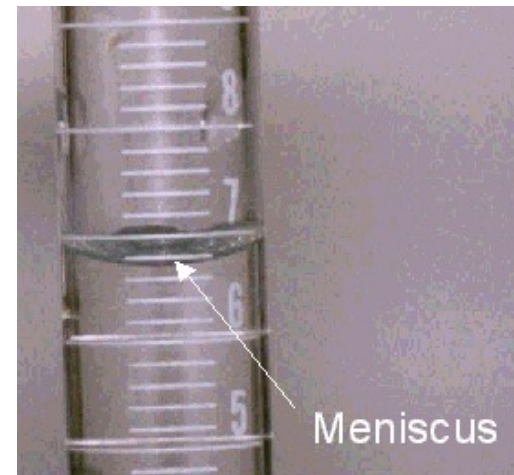
$$\theta = A\sqrt{\epsilon_r} - B$$

General Soil Calibration
Where A and B are empirical coefficients curve fit from about 20 different soil types.



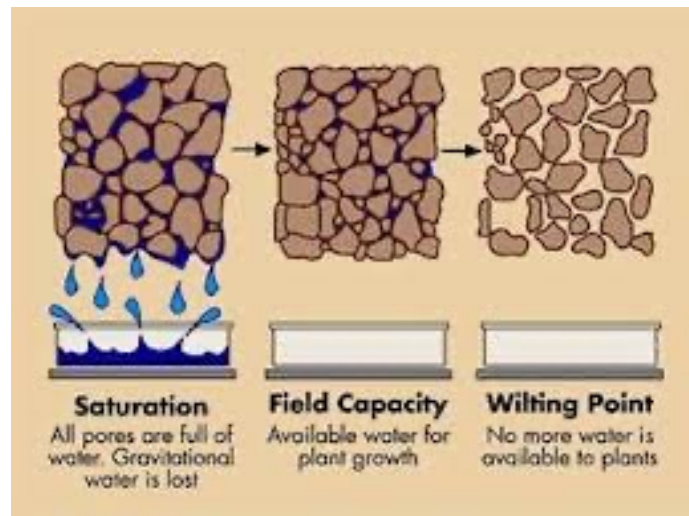
Capillary Forces

Capillary Force is a force that results from the surface tension of a liquid. The water at the surface will have a tension, meaning that the water molecules are bunched up and want to move apart. We see this if you put water in a small area. The water tension will make the water climb the walls and form a meniscus.



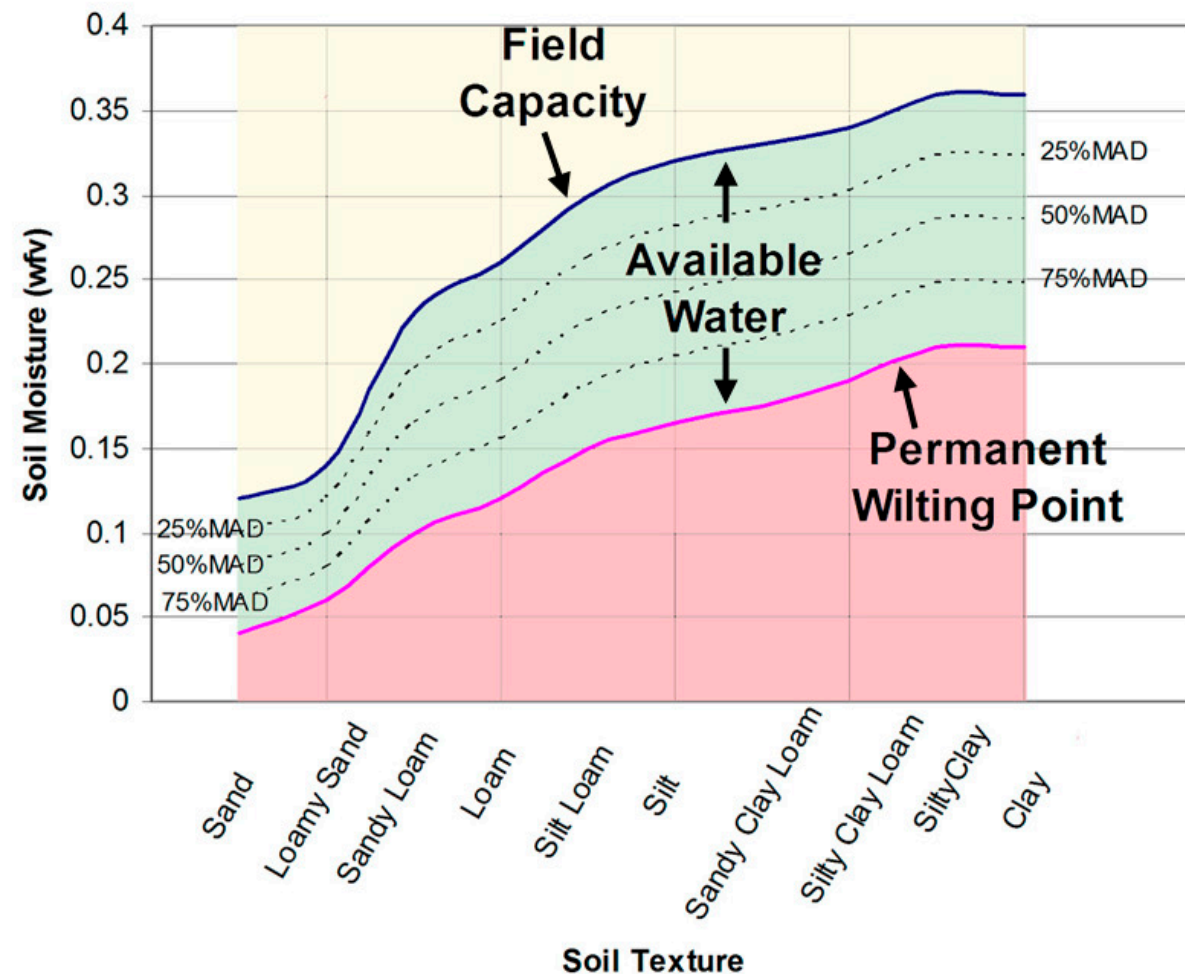
As the soil fills up with water, it will reach a point where the gravitational force will be greater than the capillary force and water will drain downward. This threshold is called field capacity

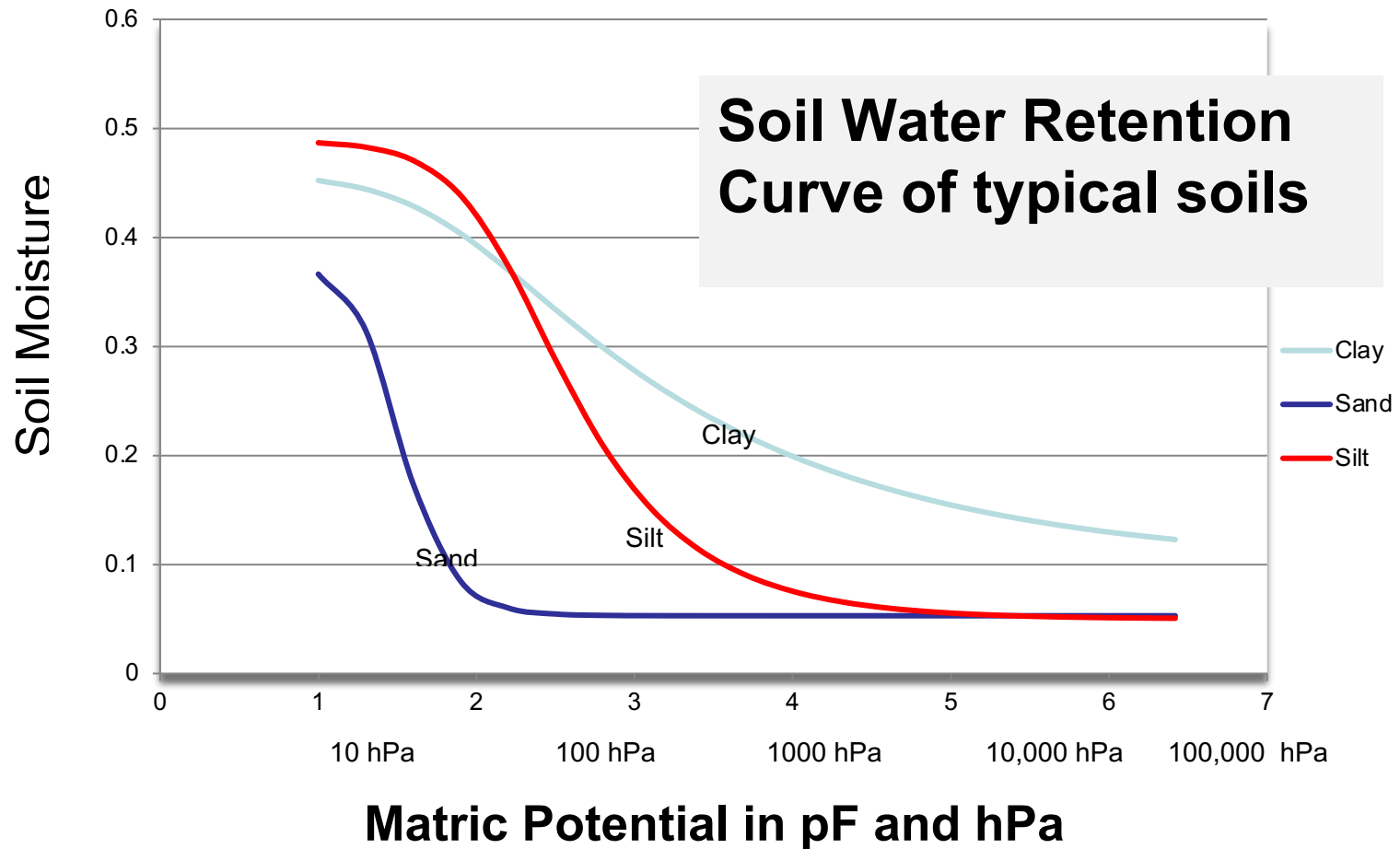
At field capacity, the water will drain out of the soil without any external pressure.



Field Capacity is a water matrix potential of 333 hPa

Soil Moisture Target

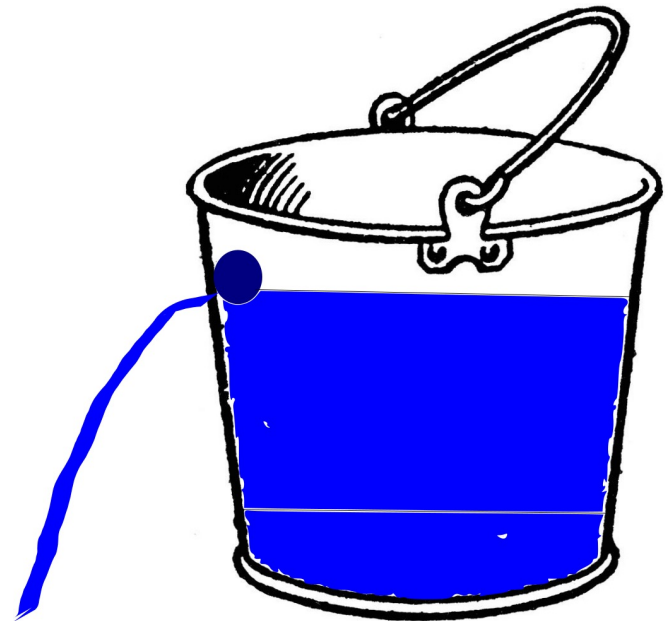
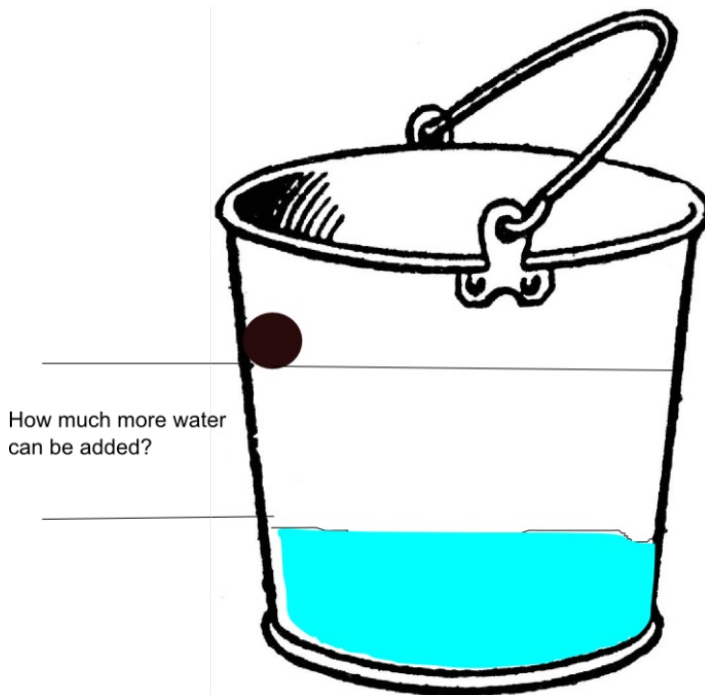




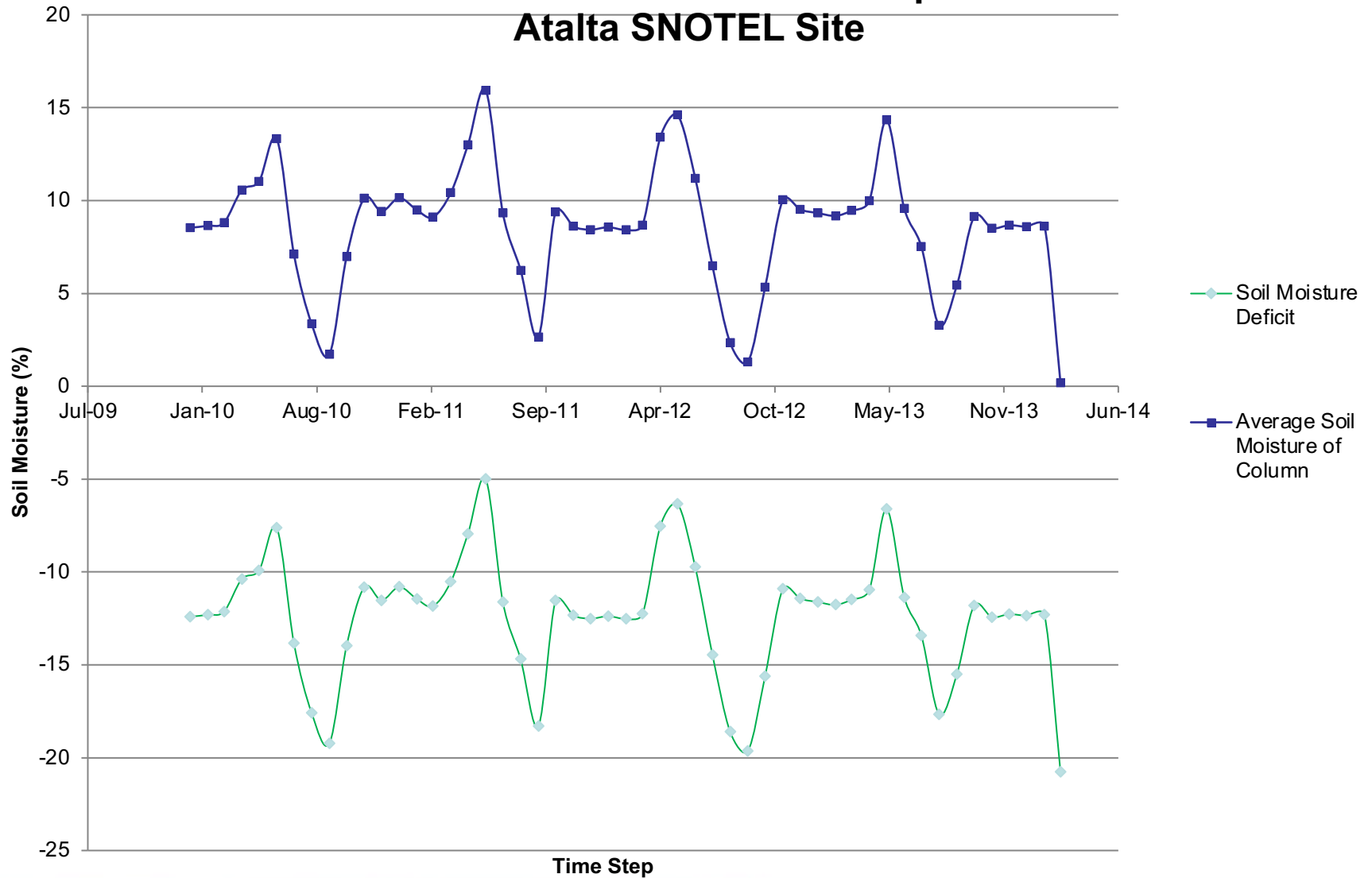
SNOTEL Site	Horizon	Top depth (cm)	Bottom Depth (cm)	Water content at 333 hPa
Atlanta Summit	A	2.00	5.00	12.79
	Bw1	5.00	23.00	24.24
	Bw2	23.00	36.00	21.44
	C	36.00	58.00	20.45
Bogus	A1	0.00	5.00	21.62
	A2	5.00	18.00	21.11
	A3	18.00	38.00	25.60
	Bw1	38.00	63.00	12.59
	Bw2	63.00	91.00	14.50
	Bw3	91.00	119.00	16.42
Jackson	A	0.00	13.00	28.36
	B1	13.00	25.00	24.53
	B2	25.00	43.00	19.00

Soil Moisture Deficit Index

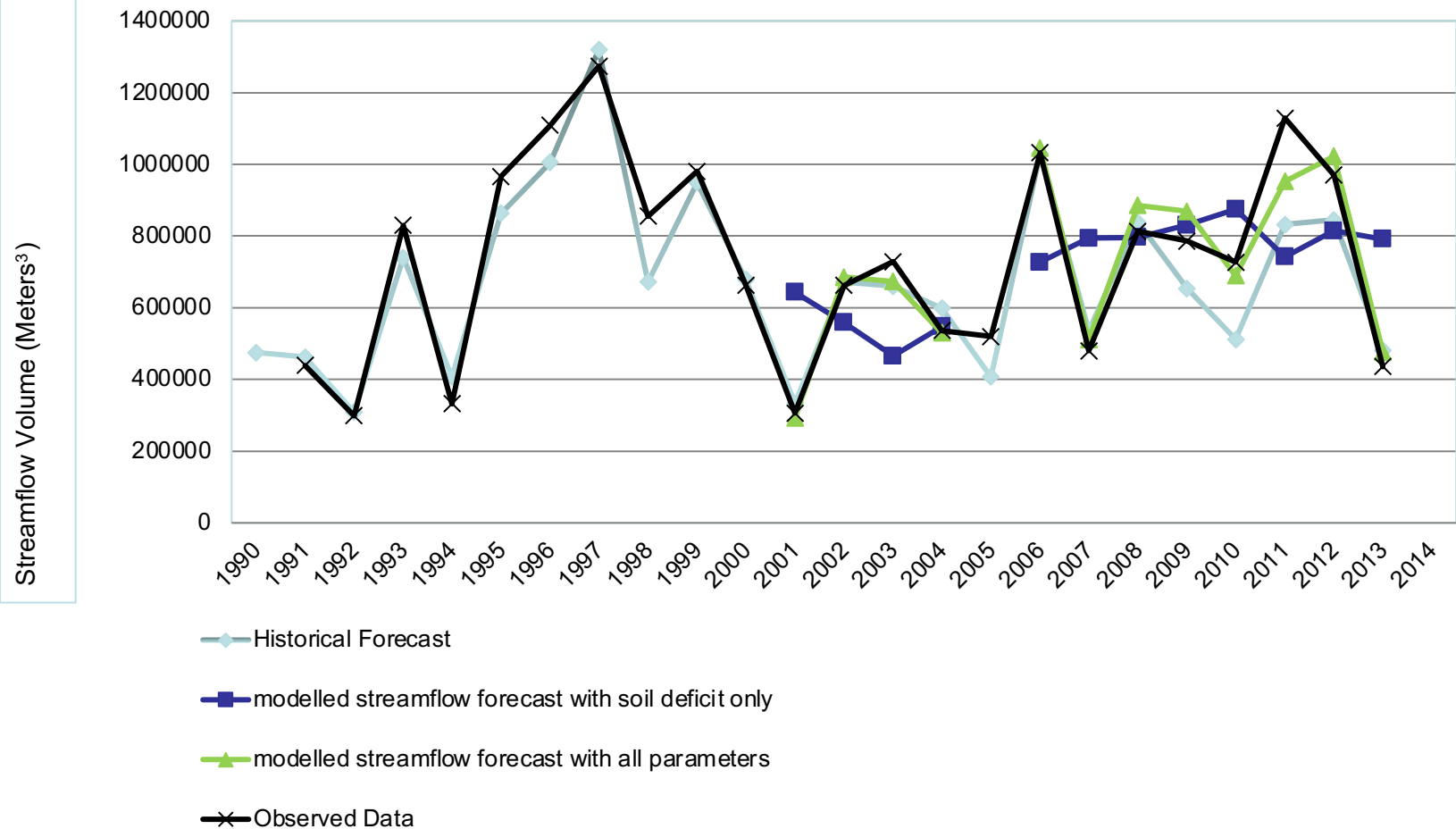
$$\theta_{di} = \bar{\theta} - \theta_{fc}$$



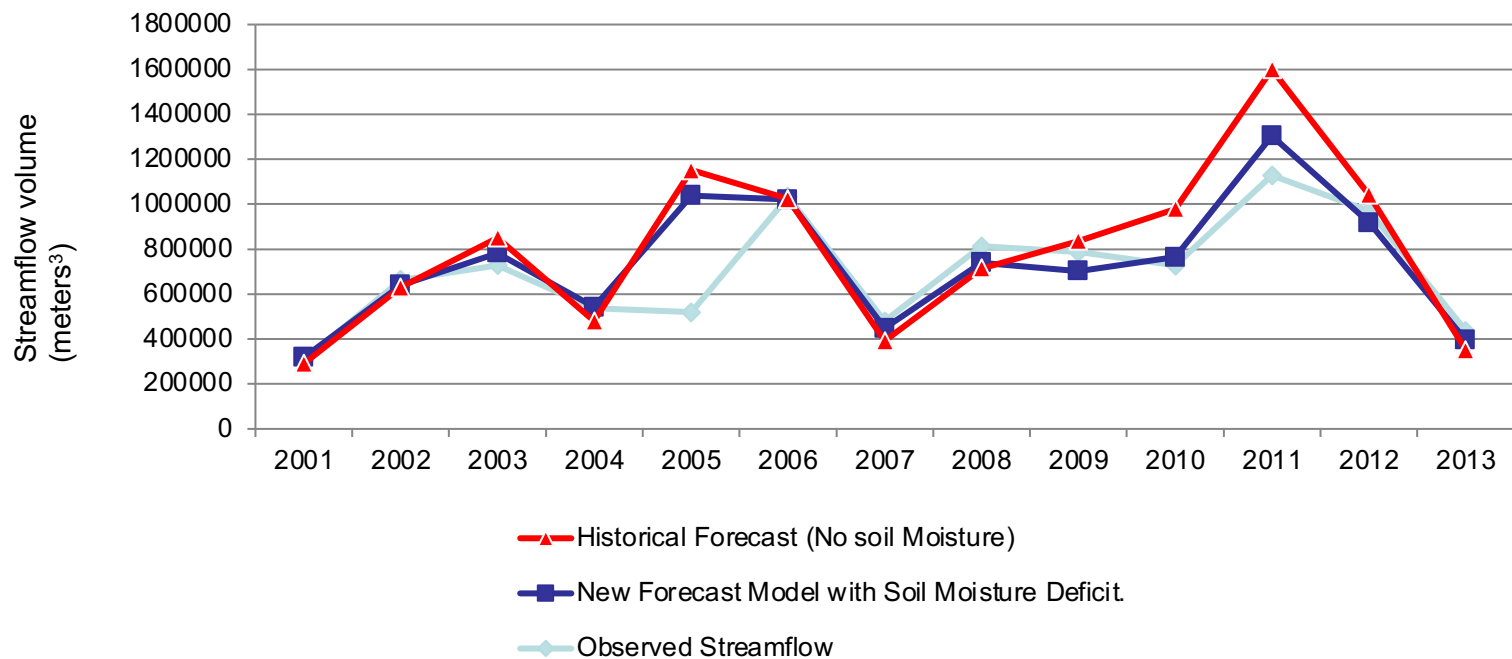
Soil Moisture vs. Time Step Atalta SNOTEL Site



The comparison of modeled streamflow forecasts to observed streamflow.



Observed streamflow at the Boise River at Twin Springs USGS gage and forecasts.



Flow in m³ per year

	Observed Data	Historical Forecast	modelled streamflow forecast with soil deficit only	modelled streamflow forecast with all parameters
2001	305579	333042	643266	293076
2002	662651	672252	558688	685276
2003	728330	659917	465745	673824
2004	536109	598242	547529	530267
2005	519541	407051		
2006	1032688	1029963	726588	1045290
2007	479784	536568	794101	511291
2008	812923	838773	795959	885850
2009	786349	653749	830922	869193
2010	726769	511898	874591	689432
2011	1128441	832605	741848	952783
2012	969946	844940	814356	1022442
2013	436132	481061	791718	476567

Percent Contribution of the soil sensors from the 3 SNOTEL sites of the forecasts.



	Atlanta Summit SNOTEL	Bogus Basin SNOTEL	Jackson Peak SNOTEL	3 Station total
Percentage of the total forecast accounted for by the soil moisture deficit for each station.	3%	5%	5%	13%

Percent Improvement over historical forecasts

Year	% Improvement Using Soil Data
2001	4.90
2002	-1.97
2003	1.91
2004	10.50
2005	
2006	-0.96
2007	5.27
2008	-5.79
2009	6.33
2010	24.43
2011	10.65
2012	7.48
2013	1.03

Improvements to the methods

- 1) Fill in the missing soil moisture data using a physical model from the meteorological data such as running HYDRUS 1D.
- 2) Used a weighted Average instead of soil column average the soil moisture def. index
- 3) Add more SNOTEL Sites, Add more soil sensors in the basin.

Conclusions/Summary

- Accurate water supply forecasts are critical.
- The soil under the snow pack represents a large reservoir for water.
- Soil moisture data measured with the Stevens Hydra Probe was modeled with meteorological parameters to forecast streamflow
- Implementation of soil moisture data significantly improved the streamflow cast

Acknowledgements

Deb Harms Soil Scientist, USDA NRCS

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Stevens Water Monitoring Systems, Inc.





Thank You

